

14 Bound States in External Fields

14.2 Radiative Corrections in External Fields

We now consider radiative corrections to the results of the previous section, due to the interaction of electrons with the quantum electromagnetic field as well as the external field of the heavy charged particles. These radiative corrections can be calculated using Feynman diagrams of the usual sort, with the whole effect of the external field being to modify the propagator of the electron field in the presence of the external electromagnetic field (and to supply the external-field-dependent renormalization counterterms shown in Eq. (14.1.1)). To be specific, the effect of inserting any number of vertices corresponding to the first term of the interaction (14.1.1) in an internal electron line of any graph is to replace the bare coordinate space propagator $-iS(x-y)$ with a corrected propagator

$$\begin{aligned} -iS_{\mathcal{A}}(x, y) \equiv & -iS(x-y) + (-i)^2 \int d^4 z_1 S(x-z_1) e\gamma^\mu \mathcal{A}_\mu(z_1) S(z_1-y) \\ & + (-i)^3 \int d^4 z_1 \int d^4 z_2 S(x-z_1) e\gamma^\mu \mathcal{A}_\mu(z_1) S(z_1-z_2) e\gamma^\nu \mathcal{A}_\nu(z_2) S(z_2-y) + \cdots \end{aligned} \quad (14.2.1)$$

where as usual

$$S(x-y) \equiv \frac{1}{(2\pi)^4} \int d^4 p \frac{-i\not{p} + m}{p^2 + m^2 - i\epsilon} e^{ip \cdot (x-y)}.$$

(We must write $S_{\mathcal{A}}$ as a function of x and y rather than of $x-y$, because the external field invalidates translation invariance.) The theorem proved in Section 6.4 tells us that Eq. (14.2.1) is the same as

$$-iS_{\mathcal{A}}(x, y) = {}_{\mathcal{A}}\langle \Omega | T\{\psi(x), \bar{\psi}(y)\} | \Omega \rangle_{\mathcal{A}}. \quad (14.2.2)$$

with the subscript $\mathcal{A}$ on the right indicating that the vacuum state $|\Omega\rangle_{\mathcal{A}}$ and electron field $\psi(x)$ are to be defined in a Heisenberg picture in which the only interaction taken into account is the interaction (14.1.1) with the external field. Inserting a complete set of intermediate states $|N\rangle_{\mathcal{A}}$ in Eq. (14.2.2) yields an expression for the propagator in terms of the Dirac wave functions u_N and v_N introduced in the previous section

$$-iS_{\mathcal{A}}(x, y) = \theta(x^0 - y^0) \sum_N u_N(x) \bar{u}_N(y) - \theta(y^0 - x^0) \sum_M v_M(x) \bar{v}_M(y). \quad (14.2.3)$$

It is also possible to obtain the propagator (14.2.2) as the solution of the *inhomogeneous Dirac equation*:

$$\left[\gamma^\mu \frac{\partial}{\partial x^\mu} + m + ie\gamma^\mu \mathcal{A}_\mu(x) \right] S_{\mathcal{A}}(x, y) = \delta^4(x-y), \quad (14.2.4)$$

which follows from the field equations (14.1.3) and anticommutation relations (14.1.7), or formally from the perturbation series (14.2.1). Also Eq. (14.2.3) tells us that the propagator satisfies boundary conditions; its Fourier decomposition contains only ‘positive frequency terms’ proportional to $\exp\{-iE(x^0 - y^0)\}$ with $E > 0$ for $x^0 - y^0 \rightarrow \infty$, and only ‘negative frequency terms’ proportional to $\exp\{+iE(x^0 - y^0)\}$ with $E > 0$ for $x^0 - y^0 \rightarrow -\infty$. The inhomogeneous Dirac equation with these boundary conditions may be used to obtain a numerical solution [4] for this propagator even in cases where the external field is too strong to allow the use of the perturbation series (14.2.1). Once the propagator $S_{\mathcal{A}}(x, y)$ has been calculated, the amplitudes for scattering in an external field can be calculated using ordinary Feynman diagrams, but with $S_{\mathcal{A}}(x, y)$ in place of $S(x-y)$ (and with $\mathcal{A}$ -dependent renormalization counterterms inserted where appropriate).

Now let us see how to use the perturbation series with this corrected propagator to calculate the shifts of bound-state energy levels. Consider the full electron propagator $S'_A(x, y)$, involving interactions of the electron with the quantum electromagnetic field as well as the external field:

$$-iS'_A(x, y) = {}_A\langle\Omega|T\{\Psi(x), \bar{\Psi}(y)\}|\Omega\rangle_A \quad (14.2.5)$$

with $\Psi(x)$ the electron field in a Heisenberg picture including all interactions, and $|\Omega\rangle$ the vacuum eigenstate of the full Hamiltonian. For a time-independent external potential we can find a complete orthonormal set $|N\rangle$ of eigenstates of the full Hamiltonian with energies E'_N . Inserting a sum over these states in the operator product in Eq. (14.2.5), we find

$$\begin{aligned} -iS'_A(x, y) &= \theta(x^0 - y^0) e^{-iE'_N(x^0 - y^0)} \sum_N U_N(\mathbf{x}) \bar{U}_N(\mathbf{y}) \\ &\quad - \theta(y^0 - x^0) e^{-iE'_N(y^0 - x^0)} \sum_N V_N(\mathbf{x}) \bar{V}_N(\mathbf{y}), \end{aligned} \quad (14.2.6)$$

where

$$\langle\Omega|\Psi(\mathbf{x}, t)|N\rangle \equiv e^{-iE'_N t} U_N(\mathbf{x}), \quad (14.2.7)$$

$$\langle N|\bar{\Psi}(\mathbf{x}, t)|\Omega\rangle \equiv e^{+iE'_N t} \bar{V}_N(\mathbf{x}). \quad (14.2.8)$$

(The sum includes an integral over continuum states as well as a sum over discrete bound states. As before, U_N and V_N are non-zero only if the state $|N\rangle$ has charge $-e$ or $+e$, respectively.) We can redefine the propagator as a function of energy rather than time

$$S'_A(\mathbf{x}, \mathbf{y}; E) \equiv \int_{-\infty}^{\infty} dx^0 e^{iE(x^0 - y^0)} S'_A(x, y). \quad (14.2.9)$$

(Time-translation invariance dictates that $S'_A(x, y)$ is a function of $x^0 - y^0$ but not of x^0 and y^0 separately.) From Eq. (14.2.6) we see that

$$S'_A(\mathbf{x}, \mathbf{y}; E) = \sum_N \frac{U_N(\mathbf{x}) \bar{U}_N(\mathbf{y})}{E'_N - E - i\epsilon} - \sum_N \frac{V_N(\mathbf{x}) \bar{V}_N(\mathbf{y})}{E'_N + E - i\epsilon}. \quad (14.2.10)$$

In particular, $S'_A(\mathbf{x}, \mathbf{y}; E)$ has a pole at any electron bound-state energy, and also at the negative of any positron bound-state energy. (Of course, positrons do not have bound states in the Coulomb field of an ordinary positively charged nucleus.)

Let us now consider the lowest-order radiative corrections to the complete propagator. The Feynman rules here give the complete propagator to this order as $S'_A = S_A + \delta S_A$, with a correction term

$$\delta S_A(x, y) = \int d^4 z \int d^4 w S_A(x, z) \Sigma_A^*(z, w) S_A(w, y), \quad (14.2.11)$$

where $i\Sigma_A^*$ is the sum of all one-loop diagrams with one incoming and one outgoing electron line (excluding final electron propagators) calculated using $S_A(x, y)$ in place of $S(x - y)$ for internal electron lines, plus second-order renormalization counterterms. Using energy variables in place of time variables, this is

$$\delta S_A(\mathbf{x}, \mathbf{y}; E) = \int d^3 z \int d^3 w S_A(\mathbf{x}, \mathbf{z}; E) \Sigma_A^*(\mathbf{z}, \mathbf{w}; E) S_A(\mathbf{w}, \mathbf{y}; E), \quad (14.2.12)$$

where

$$\Sigma_A^*(\mathbf{z}, \mathbf{w}; E) \equiv \int dz^0 e^{iE(z^0 - w^0)} \Sigma_A^*(z, w). \quad (14.2.13)$$

The effect of these radiative corrections is to change the wave functions to $U_N = u_N + \delta u_N$ and $V_N = v_N + \delta v_N$ and the bound-state energies to $E'_N = E_N + \delta E_N$, so that the complete propagator is

$$\begin{aligned} S'_A(\mathbf{x}, \mathbf{y}; E) &\simeq S_A(\mathbf{x}, \mathbf{y}; E) \\ &+ \sum_N \frac{\delta u_N(\mathbf{x}) \bar{u}_N(\mathbf{y}) + u_N(\mathbf{x}) \delta \bar{u}_N(\mathbf{y})}{E_N - E} \\ &- \sum_N \frac{\delta v_N(\mathbf{x}) \bar{v}_N(\mathbf{y}) + v_N(\mathbf{x}) \delta \bar{v}_N(\mathbf{y})}{E_N + E} \\ &- \sum_N \frac{u_N(\mathbf{x}) \bar{u}_N(\mathbf{y}) \delta E_N}{(E_N - E)^2} + \sum_N \frac{v_N(\mathbf{x}) \bar{v}_N(\mathbf{y}) \delta E_N}{(E_N + E)^2}. \end{aligned} \quad (14.2.14)$$

(We are dropping the $i\epsilon$ terms because we are now not taking E to lie in the continuum of scattering states.) We see that the shift δE_N of an electron bound-state energy is given by the coefficient of $-u_N(\mathbf{x}) \bar{u}_N(\mathbf{y}) / (E_N - E)^2$ in the complete propagator. To calculate this, we note that Eq. (14.2.3) gives

$$S_A(\mathbf{x}, \mathbf{y}; E) = \sum_N \frac{u_N(\mathbf{x}) \bar{u}_N(\mathbf{y})}{E_N - E - i\epsilon} - \sum_N \frac{v_N(\mathbf{x}) \bar{v}_N(\mathbf{y})}{E_N + E - i\epsilon}. \quad (14.2.15)$$

Inserting this in Eq. (14.2.12) gives

$$\begin{aligned} \delta S_A(\mathbf{x}, \mathbf{y}; E) &= \sum_{N,M} \frac{u_N(\mathbf{x}) \bar{u}_M(\mathbf{y})}{(E_N - E)(E_M - E)} \\ &\cdot \int d^3 z \int d^3 w \bar{u}_N(\mathbf{z}) \Sigma_A^*(\mathbf{z}, \mathbf{w}; E) u_M(\mathbf{w}) + \dots \end{aligned} \quad (14.2.16)$$

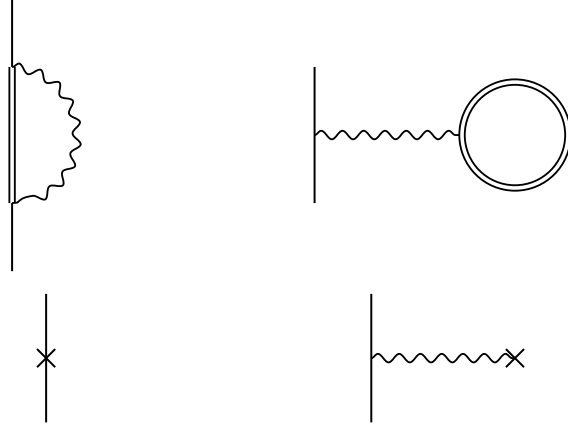


Figure 14.1: Lowest-order Feynman diagrams for the electron self-energy function $\Sigma_A^*(x, y)$ in the presence of an external field. Here double straight lines represent electron propagators S_A that include effects of the external field; single straight lines are the incoming and outgoing electron lines; wavy lines are virtual photons; the cross represents renormalization counterterms.

where the dots denote additional terms involving at least one negative-energy pole. Comparing the coefficients of $(E_N - E)^{-2}$ here and in Eq. (14.2.14), we find

$$\delta E_N = - \int d^3 x \int d^3 y \bar{u}_N(\mathbf{x}) \Sigma_A^*(\mathbf{x}, \mathbf{y}; E_N) u_N(\mathbf{y}). \quad (14.2.17)$$

The u_N are solutions of the homogeneous Dirac equation satisfying the normalization condition (14.1.8), so this is very much like ordinary first-order perturbation theory, but with $-\Sigma^*$ in place of a perturbation to the Hamiltonian.

Generally δE_N turns out to be complex. This is simply a consequence of the instability of atomic energy levels to radiative decay to lower levels; we saw in Chapter 3 that an unstable state of energy E and decay rate Γ produces poles in various amplitudes at the complex energy $E - i\Gamma/2$. The imaginary part of Eq. (14.2.17) therefore equals $-\Gamma/2$, while its real part gives the energy shift.

The Feynman diagrams for Σ^* are shown in Figure 14.1. (Note that there are new photon tadpole diagrams here because the external field breaks the Lorentz invariance and charge-conjugation invariance that forbids such diagrams in the ordinary Feynman rules.) Application of the position space Feynman rules to these diagrams gives

$$\begin{aligned} i\Sigma_{\mathcal{A}}^*(x, y) = & [e\gamma^\mu][-iS_{\mathcal{A}}(x, y)][e\gamma_\mu][-iD(x - y)] \\ & - [e\gamma^\mu\delta^4(x - y)] \int d^4z [-iD(x - z)] \text{Tr}\{[-iS_{\mathcal{A}}(z, z)][e\gamma_\mu]\} \\ & - i(Z_2 - 1)(\gamma^\mu\partial_\mu + m)\delta^4(x - y) + i\delta m\delta^4(x - y) \\ & + e\gamma^\mu(Z_2 - 1)\delta^4(x - y)\mathcal{A}_\mu(x) \\ & + i(Z_3 - 1)[e\gamma_\mu]\delta^4(x - y) \int d^4z [-iD(x - z)]\partial_\nu(\partial^\nu\mathcal{A}^\mu(z) - \partial^\mu\mathcal{A}^\nu(z)). \end{aligned} \quad (14.2.18)$$

with the renormalization constants $(Z_2 - 1)$, $(Z_3 - 1)$, and δm calculated to second order in e . (The minus sign in the second term is the usual one that accompanies closed fermion loops.)

For strong external fields with $Z\alpha$ of order unity it is necessary to calculate the configuration space electron propagator $S_{\mathcal{A}}$ and the integrals in Eqs. (14.2.17) and (14.2.18) numerically [4]. However, for weak fields we can use the first few terms of the series (14.2.1) in Eq. (14.2.18) and calculate these integrals in closed form. For this purpose, it is more convenient to work in momentum space, defining:

$$S_{\mathcal{A}}(x, y) = (2\pi)^{-4} \int d^4p' d^4p e^{ip'\cdot x} e^{-ip\cdot y} S_{\mathcal{A}}(p', p), \quad (14.2.19)$$

$$\Sigma_{\mathcal{A}}^*(x, y) = (2\pi)^{-4} \int d^4p' d^4p e^{ip'\cdot x} e^{-ip\cdot y} \Sigma_{\mathcal{A}}^*(p', p), \quad (14.2.20)$$

$$u_N(\mathbf{x}) = (2\pi)^{-3/2} \int d^3p e^{i\mathbf{p}\cdot\mathbf{x}} u_N(\mathbf{p}), \quad (14.2.21)$$

$$\mathcal{A}^\mu(x) = \int d^4q e^{iq\cdot x} \mathcal{A}^\mu(q). \quad (14.2.22)$$

(We are here committing the impropriety of using the same symbol for a function and its Fourier transform, leaving it to the displayed arguments of the functions to indicate which is which.) Then Eqs. (14.2.1) and (14.2.18) become

$$S_{\mathcal{A}}(p', p) = \frac{-i\not{p}' + m}{p'^2 + m^2 - i\epsilon} - ie \frac{-i\not{p}' + m}{p'^2 + m^2 - i\epsilon} \mathcal{A}(p' - p) \frac{-i\not{p} + m}{p^2 + m^2 - i\epsilon} + \dots \quad (14.2.23)$$

and

$$\begin{aligned} \Sigma_{\mathcal{A}}^*(p', p) = & \frac{ie^2}{(2\pi)^4} \int \frac{d^4k}{k^2 - i\epsilon} \gamma^\mu S_{\mathcal{A}}(p' - k, p - k) \gamma_\mu \\ & + [-(Z_2 - 1)(i\not{p} + m) + Z_2\delta m] \delta^4(p' - p) - ie(Z_2 - 1)\mathcal{A}(p' - p) \\ & - \frac{ie^2\gamma^\mu}{(2\pi)^4[(p - p')^2 - i\epsilon]} \int d^4q \text{Tr}\{S_{\mathcal{A}}(q, q + p' - p)\gamma_\mu\} \\ & + \frac{ie(Z_3 - 1)}{(p - p')^2 - i\epsilon} [(p - p')^2 \mathcal{A}(p' - p) - (\not{p} - \not{p}')(p - p') \cdot \mathcal{A}(p' - p)]. \end{aligned} \quad (14.2.24)$$

Because the external field is time-independent, $S_{\mathcal{A}}(x, y)$ and $\Sigma_{\mathcal{A}}^*(x, y)$ can depend on x^0 and y^0 only through the difference $x^0 - y^0$, so $S_{\mathcal{A}}(p', p)$ and $\Sigma_{\mathcal{A}}^*(p', p)$ as well as $\mathcal{A}^\mu(p' - p)$ must be proportional to $\delta(p'^0 - p^0)$:

$$\mathcal{A}^\mu(p' - p) = \delta(p'^0 - p^0) \mathcal{A}^\mu(\mathbf{p}' - \mathbf{p}), \quad (14.2.25)$$

$$S_{\mathcal{A}}(p', p) = \delta(p'^0 - p^0) S_{\mathcal{A}}(\mathbf{p}', \mathbf{p}; p^0), \quad (14.2.26)$$

$$\Sigma_{\mathcal{A}}^*(p', p) = \delta(p'^0 - p^0) \Sigma_{\mathcal{A}}^*(\mathbf{p}', \mathbf{p}; p^0). \quad (14.2.27)$$

The energy shift is then given by Eqs. (14.2.17) and (14.2.13) as

$$\delta E_N = - \int d^3 p' \int d^3 p \bar{u}_N(\mathbf{p}') \Sigma_{\mathcal{A}}^*(\mathbf{p}', \mathbf{p}; E_N) u_N(\mathbf{p}), \quad (14.2.28)$$

with $\Sigma_{\mathcal{A}}^*(\mathbf{p}', \mathbf{p}; E_N)$ given by Eqs. (14.2.23), (14.2.24), and (14.2.27). This is the master formula we shall use in the next section to calculate energy shifts in weak external fields.